

Modeling Procedure Assistance

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Task structure. Let $\mathcal{S} = \{1, \dots, N\}$ denote procedural steps and let the terminal state be **done**. A known task model G provides (i) a next-step distribution $G_{\text{next}}(s' \mid s)$ over $\mathcal{S} \cup \{\text{done}\}$, and (ii) a step-duration distribution $D_s \sim G_{\text{dur}}(\cdot \mid s)$ describing how many discrete ticks step s lasts before completion (assume its mean and variance are specified by G). To obtain Markovian tick-level dynamics, define the discrete hazard

$$h_s(\tau) := \Pr(D_s = \tau + 1 \mid D_s > \tau) = \frac{\Pr(D_s = \tau + 1)}{\Pr(D_s > \tau)}. \quad (1)$$

Latent state. The underlying (shared) latent state is

$$x_t := (s_t, \tau_t, m_t),$$

where $s_t \in \mathcal{S} \cup \{\text{done}\}$ is the current step, $\tau_t \in \{0, 1, 2, \dots\}$ is the elapsed time since entering s_t (in ticks), and $m_t = (m_{1,t}, \dots, m_{N,t})$ are reminder-memory variables (one per step).

Assistant actions. The assistant chooses

$$a_t^A \in \mathcal{A}^A := \{\text{silent}, \text{confirm}\} \cup \{\text{remind}_n\}_{n \in \mathcal{S}}.$$

Action **confirm** queries the user for step information; action **remind_n** provides guidance for step n and influences future failure probability via memory.

Human actions (interaction-only). For simplicity, the human’s main-task behavior is not modeled as a decision variable (the task progresses stochastically via G). The human only chooses an interaction action

$$i_t \in \mathcal{I} := \{\text{silent}, \text{narrate}, \text{respond}\}.$$

To avoid ambiguity, we can restrict feasible interaction actions by the assistant’s action:

$$\mathcal{I}(a_t^A) = \begin{cases} \{\text{silent}, \text{narrate}\}, & a_t^A = \text{silent}, \\ \{\text{silent}, \text{respond}\}, & a_t^A = \text{confirm}, \\ \{\text{silent}\}, & a_t^A = \text{remind}_n \text{ for some } n \in \mathcal{S}. \end{cases}$$

State transition (joint). Given (s_t, τ_t, m_t) , the task either continues in the same step or completes it stochastically each tick. Let $\chi_t \in \{0, 1\}$ indicate step completion at time t . For $s_t \in \mathcal{S}$,

$$\Pr(\chi_t = 1 \mid s_t = s, \tau_t = \tau) = h_s(\tau), \quad \Pr(\chi_t = 0 \mid s_t = s, \tau_t = \tau) = 1 - h_s(\tau). \quad (2)$$

If $\chi_t = 0$, then $s_{t+1} = s_t$ and $\tau_{t+1} = \tau_t + 1$. If $\chi_t = 1$, then $s_{t+1} \sim G_{\text{next}}(\cdot \mid s_t)$ and $\tau_{t+1} = 0$ (with **done** absorbing).

Reminder-memory updates each tick as

$$m_{n,t+1} = (1 - \lambda)m_{n,t} + \Delta_A \mathbb{I}[a_t^A = \text{remind}_n], \quad n \in \mathcal{S}, \quad (3)$$

with forgetting rate $\lambda \in (0, 1)$ and reminder increment $\Delta_A > 0$.

Failures (modeled at step completion). When a step completes ($\chi_t = 1$ and $s_t \in \mathcal{S}$), a failure indicator $e_t \in \{0, 1\}$ is sampled:

$$e_t \sim \text{Bernoulli}(f_{s_t}(m_{s_t,t})) \text{ if } \chi_t = 1, \quad e_t = 0 \text{ if } \chi_t = 0. \quad (4)$$

Here $f_n(\cdot)$ is decreasing in memory (e.g., $f_n(m) = f_n^0 \exp(-km)$ with $f_n^0 \in (0, 1)$ and $k > 0$). Each step has a severity $c_{\text{fail}}(n) \geq 0$.

Observations. The assistant receives $y_t^A = (o_t, z_t)$ each tick. Sensor observation o_t is generated by a model that can explicitly include elapsed-time information from an external step-tracking module:

$$o_t \sim P(o_t \mid s_t, \tau_t; \phi).$$

In addition, a linguistic signal $z_t \in \mathcal{Z} \cup \{\emptyset\}$ is observed; if $i_t = \text{silent}$, then $z_t = \emptyset$, otherwise

$$z_t \sim P(z_t \mid s_t, i_t; \psi).$$

Parameters ϕ, ψ capture sensor and language-channel noise.

Utility-based human interaction (bounded rationality). Let $g_t \geq 0$ denote the expected reduction in the assistant’s downstream expected failure cost at the next step completion enabled by providing an accurate disclosure of the current step at time t (defined relative to the assistant’s current belief and policy). A simple instantiation for communication choice is

$$P(i_t = \text{narrate} \mid h_t, a_t^A = \text{silent}) = \sigma(\beta(g_t - c_{\text{nar}})), \quad (5)$$

$$P(i_t = \text{respond} \mid h_t, a_t^A = \text{confirm}) = \sigma(\beta(g_t - c_{\text{resp}} - c_{\text{int}})), \quad (6)$$

with rationality $\beta > 0$ and costs $c_{\text{nar}}, c_{\text{resp}}, c_{\text{int}} \geq 0$. (When i_t is not feasible under $\mathcal{I}(a_t^A)$, its probability is defined as 0.)

Rewards. For simplicity, we assume shared reward structure, which can be maximized by minimizing the failure cost and communication cost.

$$r_t^H = -\left(c_{\text{int}}^H \mathbb{I}[a_t^A \neq \text{silent}] + c_{\text{nar}}^H \mathbb{I}[i_t = \text{narrate}] + c_{\text{resp}}^H \mathbb{I}[i_t = \text{respond}] + c_{\text{fail}}^H(s_t) e_t \mathbb{I}[\chi_t = 1]\right). \quad (7)$$

POMDP tuple (assistant; induced by the human interaction policy).

Element	Formalization	Short description
State space $\mathcal{X}$	$x_t = (s_t, \tau_t, m_t)$, with $s_t \in \mathcal{S} \cup \{\text{done}\}$, $\tau_t \in \mathbb{N}_0$, $m_t \in \mathbb{R}_{\geq 0}^N$.	Current step, time-in-step, and reminder memory (latent).
Action space $\mathcal{A}^A$	$\{\text{silent}, \text{confirm}\} \cup \{\text{remind}_n\}_{n \in \mathcal{S}}$.	Ask for state (confirm) or deliver step-specific guidance (remind_n).
Observation space Ω^A	$y_t^A = (o_t, z_t)$, $o_t \in \mathcal{O}$, $z_t \in \mathcal{Z} \cup \{\emptyset\}$.	Noisy step-tracking signal and optional speech/ASR signal.
Observation model $\mathcal{O}^A$	$o_t \sim P(o_t \mid s_t, \tau_t; \phi)$; $z_t = \emptyset$ if $i_t = \text{silent}$, else $z_t \sim P(z_t \mid s_t, i_t; \psi)$.	Sensors can include elapsed-time features; language channel is noisy.
Transition T^A	Assume a parameterized human policy $\pi_H(i_t \mid h_t, a_t^A; \theta)$ (e.g., Eqs. (5)–(6)). The induced transition is $T^A(x_{t+1} \mid x_t, a_t^A) = \sum_{i_t \in \mathcal{I}(a_t^A)} \pi_H(i_t \mid h_t, a_t^A; \theta) T(x_{t+1} \mid x_t, a_t^A, i_t),$ where T uses Eqs. (2)–(3) and G_{next} .	Task progress is stochastic via hazard; reminders (and optionally user rehearsal) update memory.
Reward R^H	Per-tick negative cost (interaction + failure), see Eq. 7.	User trades off communication burden and step failure.
Discount / termination	Episodic with absorbing terminal step done; $\gamma \in (0, 1]$ (often $\gamma = 1$).	Episode ends when the procedure reaches done.

POMDP tuple (human; interaction-only, induced by an assistant policy).

Element	Formalization	Short description
State space $\mathcal{X}$	Same latent state $x_t = (s_t, \tau_t, m_t)$.	Shared world state; the human is part of the environment for the assistant.
Action space $\mathcal{A}^H$	Interaction action $i_t \in \mathcal{I}(a_t^A)$.	Decide whether to communicate (narrate/respond) or stay silent.
Observation space Ω^H	$y_t^H = (s_t, \tau_t, a_t^A)$.	Assume the human has accurate access to current step and elapsed time, and observes the assistant's action/message.
Transition T^H	Given an assistant policy $\pi_A(a_t^A \mid h_t^A)$, the induced transition is $T^H(x_{t+1} \mid x_t, i_t) = \sum_{a_t^A \in \mathcal{A}^A} \pi_A(a_t^A \mid h_t^A) T(x_{t+1} \mid x_t, a_t^A, i_t).$	World evolves based on stochastic task progress and reminder-memory dynamics.
Reward R^H	Per-tick negative cost (interaction + failure), see Eq. 7.	User trades off communication burden and step failure.
Discount / termination	Episodic until done; $\gamma \in (0, 1]$.	Same completion condition (procedure ends at done).